

SESSION-3
MODULE-1(TASK)

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

ANS:

- **HTTP (Hypertext Transfer Protocol):** The foundational protocol used for transmitting web pages and data over the internet in plain text.
- **HTTPS (Hypertext Transfer Protocol Secure):** The encrypted version of HTTP that secures communication between a browser and a website using SSL/TLS encryption.
- **FTP (File Transfer Protocol):** A standard network protocol used specifically for transferring files between a client and a server over a network.
- **SMTP (Simple Mail Transfer Protocol):** A protocol dedicated to sending and routing outgoing emails across internet networks.

2. What is the difference between IPv4 and IPv6? Answer in 2 lines.

ANS:

IPv4 uses a 32-bit address space (expressed in decimal, e.g., 192.168.1.1) providing roughly 4.3 billion unique addresses, which are now nearly exhausted. IPv6 uses a 128-bit address space (expressed in hexadecimal, e.g., 2001:db8::ff00:42:8329), offering a virtually limitless pool of unique IP addresses to accommodate the modern explosion of internet-connected devices.

3. Run nslookup google.com on your system and write down the IP address you get.

ANS:

As an AI, I don't have a live, local operating system terminal or a static network interface to run a real-time command, but a standard DNS lookup for google.com typically resolves to its public Anycast IP addresses.

If you run **nslookup** google.com on your own machine, you will get a result that looks like this:

- **IPv4 Address:** 142.250.190.46 (or a similar variation depending on your region)
- **IPv6 Address:** 2607:f8b0:4005:805::200e

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

SESSION-3
MODULE-1(TASK)

ANS:

- **200 OK:** The request was successful, and the server delivered the expected data.
- **301 Moved Permanently:** The requested webpage has been permanently relocated to a new URL.
- **403 Forbidden:** The server understands the request, but you do not have permission to access the resource.
- **404 Not Found:** The server cannot find the requested webpage or file.
- **500 Internal Server Error:** A generic error message indicating the server encountered an unexpected condition that prevented it from fulfilling the request.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

ANS:

The Domain Name System (DNS) is structured like an inverted tree, read from right to left when resolving a web address.

Using www.google.com as our guide, here is how the hierarchy breaks down:

- **Root Level (.):** The invisible dot at the absolute end of every web address (e.g., www.google.com.). It is managed by root servers that direct queries to the correct top-level directory.
- **Top-Level Domain (TLD):** The last visible segment of the domain name. In our example, .com is the TLD, which handles commercial extensions.
- **Domain (Second-Level Domain):** The unique, custom name registered by an organization or individual. In our example, google is the domain.
- **Subdomain:** A prefix added to the main domain to organize distinct sections of a website. In our example, www (World Wide Web) serves as the subdomain directing users to the primary web server.